



Course Advanced Topics in Mathematics (2024-2025)

Code / Version MATH73235 (101)

Total Hours 56

Credits 4

PreRequisite(s) MATH73050 (100) Probability & Statistics
or MATH73050 (101) Probability & Statistics
or MATH73050 (102) Probability & Statistics

CoRequisite(s)

Course Description

This course is divided between advanced topics selected from calculus, statistics and discrete mathematics. Topics are selected to support advanced studies in natural sciences and engineering. Commonly selected topics would include aspects of partial derivatives, surface integrals, spherical and cylindrical coordinates, methods of proof, number and set theory, stochastic processes and queuing theory.

PLAR Eligible: Yes

Eligible for Supplemental: Yes (See eligibility requirements in the program handbook)

Resources

eText Program | eText | Erwin Kreyszig. Advanced Engineering Mathematics (10). Wiley. ISBN: 9781119446842.

Program Outcomes Supported

1094C - B.Eng - ESE (Hon) (2601) - Level 8

PLO-1: Apply university level mathematics, natural sciences, and engineering principles to solve engineering problems involving electronic systems.

Course Outcomes

Successful completion of this course will enable the student to:

1. Represent and solve problems in cylindrical and spherical coordinate systems [GA: Problem Analysis]
2. Model physical phenomena using Partial Differential Equations (PDEs), such as string vibration, membrane vibration, wave propagation and heat transfer [GA: Problem Analysis]
3. Solve PDEs using various methods, including Fourier and Laplace transforms [GA: Problem Analysis]
4. Perform operations on vectors in 2D and 3D Cartesian and curvilinear spaces [GA: Problem Analysis]
5. Compute properties of vector functions [GA: Problem Analysis]
6. Define and compute double and triple integrals over general regions [GA: Problem Analysis]
7. Compute line and surface integrals [GA: Problem Analysis]

Unit Outcomes

Successful completion of the following units will enable the student to:

1.0 Coordinate Systems

- 1.1 Represent points and vectors in Cartesian, Cylindrical, and Spherical 3D spaces. [GA: PA1 Decomposition]
- 1.2 Convert between Cartesian and Cylindrical coordinates [GA: PA1 Decomposition].
- 1.3 Convert between Cartesian and Spherical coordinates [GA: PA1 Decomposition].



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2.0 Partial Differential Equations (PDEs)

- 2.1 Discuss concepts and applications of PDEs [GA: KB2 Concepts]
- 2.2 Model a vibrating string using the Wave Equation [GA: KB2 Concepts].
- 2.3 Solve the wave equation using the Fourier Series [GA: PA1 Decomposition]
- 2.4 Solve the wave equation using D'Alembert's method [GA: PA2 Methodology]
- 2.5 Model a vibrating rectangular membrane using the two-Dimensional Wave Equation [GA: KB2 Concepts]
- 2.6 Solve two-dimensional wave equations using Double Fourier Series [GA: PA1 Decomposition]
- 2.7 Model heat flow using the Heat Equation [GA: KB2 Concepts]
- 2.8 Solve the heat equation using the Fourier Series. [GA: PA2 Methodology]
- 2.9 Solve the steady state two-dimensional heat problem [GA: PA1 Decomposition]
- 2.10 Solve the heat transfer equation along a bar using Fourier Integrals and Transforms [GA: PA1 Decomposition]
- 2.11 Discuss Laplace's Equation in Cartesian, Cylindrical and Spherical Coordinates [GA: KB2 Concepts]
- 2.12 Use Laplacian in Polar Coordinates to solve a circular membrane problem [GA: PA2 Methodology]
- 2.13 Solve PDEs using the Laplace Transforms [GA: PA2 Methodology]

3.0 Vector Calculus

- 3.1 Discuss the properties, calculation, and applications of the gradient, divergence, and curl. [GA: KB2 Concepts]
- 3.2 Evaluate line and surface integrals [GA: PA1 Decomposition]
- 3.3 Discuss the properties of line and surface integrals [GA: KB2 Concepts]
- 3.4 State Green's Theorem and use it to evaluate path integrals [GA: PA2 Methodology]
- 3.5 State Gauss's theorem (the Divergence Theorem) and use it to evaluate surface and volume integrals [GA: PA2 Methodology]
- 3.6 State Stoke's Theorem and use it to evaluate surface integrals [GA: PA2 Methodology].

Evaluation

The minimum passing grade for this course is 60 (C).

In order to successfully complete this course, the student is required to meet the following evaluation criteria:

Description	Quantity	Percentage
1. Mid-Term Exam(s)	1	35.00 %
2. Final Exam	1	35.00 %
3. Quiz(zes)	6	30.00 %
		100.00 %

Evaluation Mapping

1. Represent and solve problems in cylindrical and spherical coordinate systems [GA: Problem Analysis]

Final Exam (35.00%), Quantity: 1

Quiz(zes) (30.00%), Quantity: 6

2. Model physical phenomena using Partial Differential Equations (PDEs), such as string vibration, membrane vibration, wave propagation and heat transfer [GA: Problem Analysis]

Mid-Term Exam(s) (35.00%), Quantity: 1

Final Exam (35.00%), Quantity: 1



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Quiz(zes) (30.00%), Quantity: 6

3. Solve PDEs using various methods, including Fourier and Laplace transforms [GA: Problem Analysis]

Mid-Term Exam(s) (35.00%), Quantity: 1

Quiz(zes) (30.00%), Quantity: 6

4. Perform operations on vectors in 2D and 3D Cartesian and curvilinear spaces [GA: Problem Analysis]

Final Exam (35.00%), Quantity: 1

Quiz(zes) (30.00%), Quantity: 6

5. Compute properties of vector functions [GA: Problem Analysis]

Final Exam (35.00%), Quantity: 1

Quiz(zes) (30.00%), Quantity: 6

6. Define and compute double and triple integrals over general regions [GA: Problem Analysis]

Mid-Term Exam(s) (35.00%), Quantity: 1

Quiz(zes) (30.00%), Quantity: 6

7. Compute line and surface integrals [GA: Problem Analysis]

Final Exam (35.00%), Quantity: 1

Quiz(zes) (30.00%), Quantity: 6

Web-based Tools

This course may be using web-based services with data centres outside of Canada. Students may be expected to complete assessments where information is transmitted outside of Canada. Students who do not wish to submit their information to other countries have the right to opt-out. It is the responsibility of the student to notify the instructor in the first week of term if they have any concerns. The alternative may require the student to attend the campus testing centre at a designated time.

Notes

This course may require you to use generative AI. Check the Instructional Plan to confirm where and how AI use is required. If there is no reference to AI in the Instructional Plan, it is not permitted in this course. When not permitted, any use of AI to complete course assessments or other activities is considered a violation of the College's academic integrity policy.

Software to Detect Academic Offences: Software may be used to screen assignments or invigilate exams in this course. It is the responsibility of the professor to notify students of any such technologies in advance of their use. It is the responsibility of the student to notify the professor if they, at the time assignment details are provided, wish to opt-out and be provided with an alternate process.

Academic integrity is expected and required of all Conestoga students. It is a student's responsibility to maintain compliance with Conestoga's Academic Integrity Policy at all times.

Conestoga College is committed to providing academic accommodations for students with documented disabilities.

An Instructional Plan will be available at the beginning of the course and will be referred to in conjunction with this course outline.

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School Engineering & Technology



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